

Software Engineering Lab 2

Agile Backlog Creation & Sprint Simulation in Jira

Academic Elective Bidding & Allocation System

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This report converts the corrected Lab 1 requirements for Problem Statement #05 into a Jira-style backlog and one-week Scrum sprint simulation.

1. Epics

Epic	Description
EPIC-1: Student Bidding and Preferences	View eligible electives, rank choices, and allocate bidding credits.
EPIC-2: Eligibility and Allocation Engine	Validate prerequisites and allocate seats using capacity and timetable constraints.
EPIC-3: Registrar Configuration and Oversight	Configure offerings and review allocation outcomes.
EPIC-4: Results and Notifications	Publish allocation results, waitlists, exceptions, and notifications.

2. User Stories and Estimates

ID	Epic	Summary	Priority	Points
US-001	EPIC-1	View eligible electives	High	3
US-002	EPIC-1	Distribute 100 bidding credits	High	5
US-003	EPIC-1	Save and submit ranked preferences	High	5
US-004	EPIC-2	Validate allocation constraints	High	5
US-005	EPIC-2	Run elective allocation	High	5
US-006	EPIC-3	Configure offerings and rules	Medium	3
US-007	EPIC-4	View allocation result	Medium	3
US-008	EPIC-4	Review audit and conflict views	Low	3

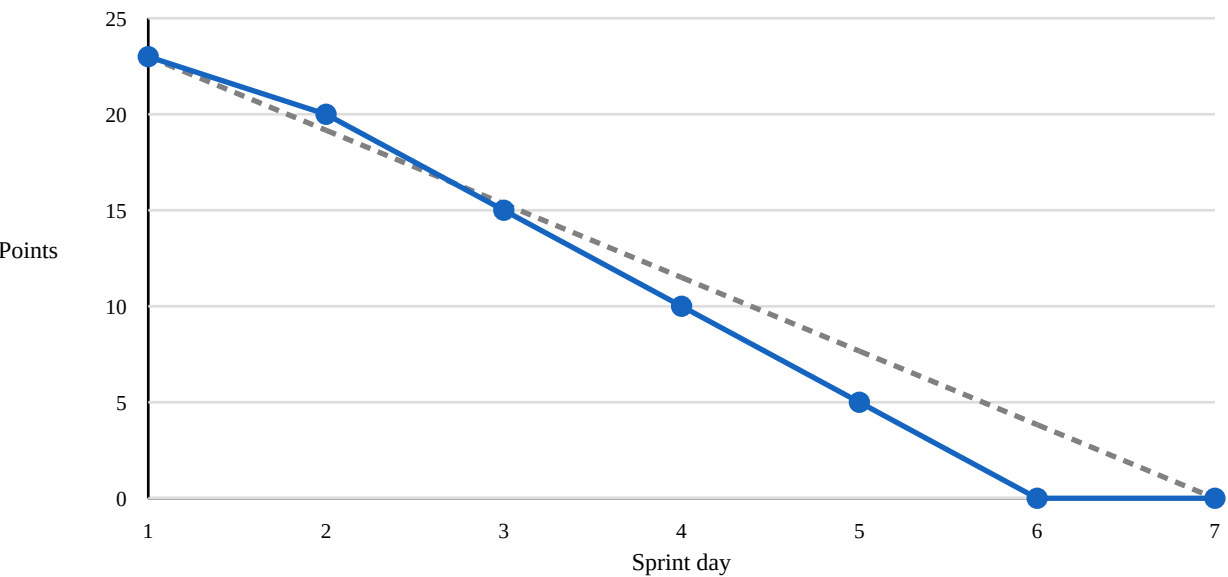
3. Sprint 1 Simulation

Sprint duration: one week. Select US-001 through US-005 for a commitment of **23 story points**. Keep US-006 through US-008 in the backlog.

Day	Completed work	Remaining
Day 1	Sprint started; selected stories To Do	23
Day 2	US-001	20
Day 3	US-002	15
Day 4	US-003	10
Day 5	US-004	5
Day 6	US-005; sprint completed	0
Day 7	Buffer / review	0

4. Burndown Chart

Sprint 1 Burndown Chart



5. Reflection

Did the estimates reflect the actual effort?

The estimates reflected relative complexity. Viewing eligible electives was a focused 3-point story, while credit validation, preference submission, constraint validation, and allocation were 5-point stories because they combine business rules and edge cases.

Was the backlog well-prioritized?

Yes. The sprint prioritizes the student-to-allocation path. Registrar configuration and reporting remain in the backlog because they support the core flow but are not prerequisites for demonstrating allocation.

How did the simulated sprint align with your plan?

Sprint 1 commits to 23 points across five stories. The order follows the dependency chain from eligible electives through credit validation, preference submission, constraint checking, and final allocation.

What insights did the burndown chart give about team capacity?

The burndown shows that a 23-point commitment is achievable within one week in the simulation. The three unselected stories should be reserved for a later sprint unless additional capacity is available.

6. Jira Evidence Checklist

- Backlog screenshot with epics and stories.
- Story point and priority evidence.
- Active Sprint board screenshot.
- Burndown Chart screenshot.
- This PDF uploaded to GitHub.